Random Variables

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Introduction

A **random variable** is a function that assigns a real number to each outcome of a random experiment. The concept packages probabilistic thinking into the language of numbers – means, variances, correlations – and is the abstraction on which every statistical method rests.

Prerequisites

Sample spaces, events, and the probability measure.

Theory

Given a probability space $(\Omega, \mathcal{F}, P)$, a **random variable** is a measurable function

$$X : \Omega \to \mathbb{R}$$

such that $\{X \leq x\} = \{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. Measurability ensures that we can assign probabilities to events defined in terms of X.

The **distribution** of X is the probability measure it induces on $\mathbb{R}$:

$$P_X(A) = P(X \in A) = P(\{\omega : X(\omega) \in A\}).$$

In practice, we rarely work with Ω directly; we work with the distribution P_X, summarised by its CDF $F_X(x) = P(X \leq x)$ and (when it exists) its PMF p_X or density f_X.

Random variables are classified as:

- **Discrete**: takes values in a countable set. Described by a PMF.
- **Continuous** (absolutely continuous): has a density. Described by a PDF.
- **Mixed**: combines discrete atoms and a continuous part. Common for censored / thresholded data.

Transformations of random variables give new random variables: if X is a random variable and $g : \mathbb{R} \to \mathbb{R}$ is measurable, then $Y = g(X)$ is a random variable with distribution determined by X and g.

Assumptions

A random variable is well-defined provided the underlying probability space exists and the function is measurable. Measurability is automatic for every function that arises in practice (continuous or piecewise-continuous).

R Implementation

In R, we usually work with random variables via their distributions.

```r
set.seed(2026)

# Discrete random variable: binomial
X_discrete <- rbinom(10000, size = 10, prob = 0.3)
head(sort(unique(X_discrete)))
prop.table(table(X_discrete))[1:5]      # empirical PMF

# Continuous random variable: normal
X_continuous <- rnorm(10000, mean = 0, sd = 1)
quantile(X_continuous, c(0.025, 0.5, 0.975))

# Transformation: Y = g(X) = X^2 of a standard normal
Y <- X_continuous^2
summary(Y)
var(Y)     # theoretical variance is 2 (Y ~ chi-squared with 1 df, variance 2)

# Mixed RV: censored at 0 (negative values truncated)
X_mixed <- pmax(rnorm(10000), 0)
mean(X_mixed == 0)
mean(X_mixed)
```

Output & Results

```
X_discrete       0       1       2       3       4
empirical   0.0306  0.1195  0.2314  0.2643  0.2059

         2.5%          50%         97.5%
   -1.935       -0.0089      1.9689

       Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
      0.000    0.128     0.455     1.008    1.319    17.59
[1] 2.078                                              # var(X^2)

[1] 0.499                                              # P(X = 0)
[1] 0.399                                              # mean of censored
```

Each quantity (PMF, quantile, variance of a transformation, mixed distribution's atom) is a statistic of the corresponding random variable.

Interpretation

Random variables let us move from "outcome space" thinking to "number space" thinking. A methods section typically declares the random variable implicitly: "Let X be the systolic blood pressure of a randomly selected patient" specifies the random variable without bothering with the underlying Ω.

Practical Tips

- For most applied work, think of random variables as "generators of data" with a specified distribution; the measurability formalism is background.
- A random variable is distinct from its realisation. "X" is the variable; "x_i" is the observed value for unit i.
- Vector-valued random variables (**random vectors**) generalise the scalar case; their distribution is the joint distribution of the components.

- Transformations preserve measurability; common operations (sums, products, functions) are always measurable in practice.
- In R, `d*`, `p*`, `q*`, `r*` families of functions parameterise distributions; think of them as computing quantities of random variables with the specified distribution.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots